WDT-Bench: A Word Deletion Benchmark for Implicitly Evaluating Syntactic Competence of Large Language Models

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***Abstract*—Syntactic competence is a core feature of human language and a key aspect when evaluating large language models (LLMs). Recently, the Word Deletion Task was proposed to implicitly test whether LLMs actively construct syntactic structures during language processing. Based on this task, we present WDT-Bench, an open-source benchmark that standardizes the task into a systematic syntactic evaluation comprising six tests. Using natural sentences and carefully constructed diagnostic sentences, WDT-Bench implicitly assesses multiple dimensions of syntactic competence, including general constituent-based processing, cross-linguistic generalization, and syntax-semantics interaction. We evaluate LLMs across model size (Qwen Flash to Qwen Max) and architecture (Qwen Max, DeepSeek V4, GPT-5.5, Claude 4.7, Gemini 3.1, Grok 4.20). The results show that (1) LLMs exhibit constituent-based processing that scales with model size, whereas small models such as Qwen Flash cannot reliably identify constituents; (2) a consistent cross-linguistic divergence holds across all LLMs, with distinct syntactic strategies adopted for Chinese and English; and (3) for most LLMs, syntactic processing depends on semantic content, whereas DeepSeek V4 remains substantially less dependent on semantic content. WDT-Bench and its supplementary appendices are publicly available at https://github.com/Aprde/WDT-Bench.**

***Index Terms*—large language models, syntactic competence, evaluation, benchmark**

I. Introduction

Classic linguistic theory holds that humans combine words into complex, hierarchically nested syntactic structures during language processing, and this syntactic competence is considered a core feature of human language [1], [2]. A central question in the era of large language models (LLMs) is whether these models also construct such syntactic structures for language comprehension [3], [4]. Despite extensive investigation, this question remains unresolved: while some studies show that LLMs exhibit human-like syntactic behavior in tasks that explicitly test grammar [5], [6], others argue that strong performance on these tasks can be explained by superficial statistical cues, without requiring genuine syntactic knowledge [7]. To address this, we present WDT-Bench, an open-source benchmark based on the Word Deletion Task [8], which evaluates syntactic competence of LLMs through implicit behavioral measures.

A growing body of work has attempted to evaluate the syntactic competence of LLMs. Grammaticality judgment tasks, such as those in BLiMP [5] and SyntaxGym [6], test whether LLMs can distinguish grammatical from ungrammatical sentences. Probing methods train classifiers on model representations to decode syntactic information [9]. However, these approaches share critical limitations. Performance on grammatical tasks does not entail that LLMs deploy syntactic structures during natural language processing [7]. Moreover, these benchmarks involve common task formats that LLMs may have encountered extensively during pretraining, raising concerns about data leakage and overfitting. Probing methods, on the other hand, demonstrate that syntactic information is encoded in model representations, but cannot establish that this information is used for language comprehension [10].

Recently, Liu et al. [8] introduced the word deletion task, a novel task that tests whether LLMs actively construct syntactic structures during language processing. In this task, a model is presented with a demonstration in which a word string has been removed from a complete sentence, and the model is required to infer the underlying rule and apply it to a new test sentence. Since only a single demonstration is provided, the task is highly underspecified—many rules are compatible with the demonstration (e.g., deleting by position or deleting by length). Although the task makes no reference to syntax, Liu et al. found that both humans and LLMs prefer to delete constituents (i.e., syntactic units such as noun phrases), suggesting that they use syntactic structure to perform the task. Since the task elicits syntactic behavior without any instruction, it serves as an implicit measure of syntactic competence [8], [11]. Crucially, the task operates under out-of-distribution conditions: LLMs must generalize to an unfamiliar task, making it robust to data leakage and superficial statistical cues. However, Liu et al.'s analyses were conducted primarily on ChatGPT with copyrighted materials that cannot be publicly distributed, preventing large-scale reproducible evaluation across model size and architecture.

In this paper, we standardize the word deletion task into WDT-Bench, an open-source, multi-dimensional benchmark for syntactic evaluation. WDT-Bench comprises two complementary components: (1) General Tests (Tests 1–3), based on natural sentences, enabling large-scale evaluation of general constituent recognition, constituent preference, and constituent localization; and (2) Diagnostic Tests (Tests 4–6), using carefully constructed sentences targeting cross-linguistic generalization, semantic independence of syntactic processing, and syntax-semantics integration. We evaluate LLMs ranging from Qwen Flash to Qwen Max, as well as DeepSeek V4, GPT-5.5, Claude 4.7, Gemini 3.1, and Grok 4.20, providing a systematic cross-model comparison on syntactic competence.

Our contributions are as follows:

1) We introduce WDT-Bench, an open-source benchmark that standardizes the word deletion task for implicit evaluation of syntactic competence.
2) We conduct a systematic evaluation across LLMs of different model sizes and architectures, examining the effects of model size, model architecture, prompt design, and the number of in-context demonstrations.
3) We find that LLMs exhibit constituent-based processing that scales with model size, a consistent cross-linguistic divergence in syntactic strategies, and syntactic processing that generally interacts with semantics (with DeepSeek V4 as an exception).

II. WDT-Bench

A. Task Description

The word deletion task is a one-shot learning task designed to test the active use of syntactic structures in LLMs [8]. Each trial of the task consists of a demonstration (a complete sentence paired with a version that has one constituent removed) and a test sentence. The task instruction makes no reference to syntax: the model is simply asked to apply the same transform to the test sentence. If a model tends to delete a single complete constituent, it is taken as evidence that the model actively employs syntactic structure in language processing [8]. Liu et al. [8] validated this interpretation, ruling out non-syntactic strategies such as deletion based on position or string length. Therefore, the proportion of single-constituent deletions serves as an implicit measure of syntactic competence. The full prompt of the word deletion task is shown in Appendix Table S2.

We adopt the following terminology throughout this paper. A constituency tree is a hierarchical representation of syntactic structure in which each internal node corresponds to a syntactic category (e.g., noun phrase, NP; verb phrase, VP; prepositional phrase, PP). A constituent is any non-terminal node of this tree. The node category of a constituent refers to its syntactic category (e.g., NP). The parent category refers to the syntactic category of the node immediately dominating it in the tree (e.g., if an NP is dominated by a PP, its parent category is PP).

TABLE I
EXAMPLES OF TESTS 1–3. CONSTRUCTION PROCEDURE FOR EACH TEST IS SHOWN IN GRAY.

	Demonstration	Test sentence
Test 1	Random sentence, random constituent deleted *Many of ~~his peers~~ feel the same way*	Different random sentence *There are few factories and no mines*
Test 2	NP-VP-NP sentence, second NP deleted *Estimated volume was ~~a light 2.4 million ounces~~*	NP-VP-PP-NP sentence *Only a few books fell in the reading room*
Test 3	Random sentence, random constituent deleted *That brought ~~a chain reaction~~ in the industry*	Same structure as demonstration *Other furriers have also placed more weight on retailing*

B. Benchmark Construction

WDT-Bench comprises six tests: general tests (Tests 1–3) for large-scale evaluation and diagnostic tests (Tests 4–6) for targeted analysis. Since evaluating deletion behavior requires determining whether LLMs tend to delete constituents, reference constituency trees serve as ground truth. For the general tests, reference constituency trees are derived from the CoNLL-2000 shared task dataset [12] (See Appendix A for detailed tree construction). For the diagnostic tests, reference constituency trees are manually constructed.

Sentence filtering. We apply the following exclusion criteria on the CoNLL-2000 dataset: we remove sentences containing named entities or sentence-internal punctuation and retain only sentences with 3-15 words and a syntactic depth between 3 and 8. The statistics of the sentences used in our study are summarized in Appendix Table S1.

General tests. The general tests comprise three tests, each examining a different aspect of the model's syntactic competence: whether the model identifies and deletes a complete constituent (Test 1), which constituent-based deletion rule it prefers (Test 2), and whether it can locate the structurally corresponding constituent in the test sentence (Test 3). The three tests differ in how the demonstration and test sentences are constructed (see Table I), as described below.

Test 1 (Constituent Recognition). In Test 1, a sentence in CoNLL-2000 is randomly selected as the demonstration, with a randomly chosen constituent removed. A different sentence is selected as the test sentence. This tests whether the model prefers to delete constituents in the test sentence. Each model response is classified into one of five categories: (1) the deleted string is a single complete constituent; (2) the deleted string spans multiple complete constituents; (3) the deleted string is a substring within a single constituent; (4) the deleted string is one complete constituent plus part of an adjacent constituent; (5) other, including cases where the model returned the original sentence unchanged or deleted a non-contiguous word string. We measure the proportion of each category. The proportion of single-constituent deletions is referred to as the constituent rate.

Test 2 (Constituent Preference). In Test 2, the demonstration uses an NP-VP-NP sentence with the second NP deleted. The test sentence has an NP-VP-PP-NP structure. This tests which deletion rule the model applies in the task. We consider

TABLE II
EXAMPLES OF TESTS 4–6. CONSTRUCTION PROCEDURE FOR EACH TEST IS SHOWN IN GRAY.

	Demonstration	Test sentence
Test 4	NP-VP-NP parallel sentence, second NP deleted *The old man feeds ~~the stray cat~~* *那个老人 喂 ~~那只流浪猫~~*	NP-VP-PP-NP parallel sentence *The accident happened on Saturday night* *这场意外 发生 在 周六 晚上*
Test 5	NP-VP-NP parallel sentence, second NP deleted *The old man feeds ~~the stray cat~~* *那个老人 喂 ~~那只流浪猫~~*	NP-VP-PP-NP nonsense sentence *The construction happened on giant economy* *这屏幕发生在驾照精神*
Test 6	NP-VP-NP sentence, second NP deleted *The photographer adjusted ~~the lens~~*	PP attachment ambiguity **Structure 1** *The guy caught the rat with a scar* **Structure 2** *The guy caught the rat with a trap*
	NP-of-NP-VP sentence, second NP deleted *The plains of ~~the tribe~~ looked strange*	Adjunct attachment ambiguity **Structure 1** *The letter of the writer that had blonde hair arrived this morning* **Structure 2** *The writer of the letter that had blonde hair arrived this morning*

two possible rules: node-category rule and parent-category rule [8]. The node-category rule predicts that the model deletes a constituent whose node category matches that of the deleted constituent in the demonstration: since an NP was deleted in the demonstration, the model deletes an NP in the test sentence (i.e., the NP under the PP). The parent-category rule predicts that the model deletes a constituent whose parent category matches that of the deleted constituent in the demonstration: since the deleted NP was governed by VP in the demonstration, the model deletes the constituent governed by VP in the test sentence (i.e., PP). We measure the explained ratio, i.e., the proportion of model responses that can be explained by node-/parent-category rule.

Test 3 (Constituent Localization). While Test 1 assesses whether the model deletes any constituent, Test 3 assesses whether the model can locate the specific constituent that structurally corresponds to the one deleted in the demonstration. In Test 3, two sentences with identical syntactic structure (e.g., both are NP-VP-NP-PP-NP sentences in Table I) are selected; one serves as the demonstration, with a randomly chosen constituent deleted; the other as the test sentence. Since the syntactic structures are identical, there exists a uniquely correct corresponding constituent in the test sentence. We measure the proportion of trials in which the model deletes exactly the corresponding constituent.

Diagnostic Tests. The diagnostic tests use carefully constructed sentences to probe specific dimensions of syntactic competence [8], including cross-linguistic generalization (Test 4), semantic independence (Test 5), and syntax-semantics integration (Test 6). The examples of diagnostic tests are shown in Table II.

Test 4 (Cross-linguistic Generalization). In Test 4, Chinese-English parallel sentences serve as demonstration-test pairs. The parallel sentences share identical syntactic structure, word order, and semantic content across Chinese and English. The demonstration uses an NP-VP-NP sentence with the second NP deleted; the test sentence has an NP-VP-PP-NP structure. This tests whether the model prefers to delete constituents and applies the same deletion rule (node-category rule vs. parent-category rule) across languages. We measure the constituent rate and explained ratio for both languages.

Test 5 (Semantic Independence). In Test 5, nonsense sentences (syntactically well-formed but meaningless) are used as test sentences. The nonsense sentences are constructed from the NP-VP-PP-NP parallel sentences used in Test 4 by replacing content words (except verbs) with random words of the same part of speech, preserving syntactic structure while disrupting semantic coherence. The demonstration uses the meaningful NP-VP-NP parallel sentence with the second NP deleted. This tests whether the model can solely rely on the syntactic structure to identify constituents, or whether semantic content is also required. We measure the constituent rate and explained ratio for nonsense sentences and compare with the results of Test 4, to quantify the degree to which syntactic behavior depends on semantic content.

Test 6 (Syntax-semantics Integration). In Test 6, two types of syntactically ambiguous sentences are used as test sentences: (1) PP attachment ambiguity: sentences of the form "NP1 verb NP2 PP", where the PP can attach to either NP2 (structure 1) or the verb (structure 2). (2) Adjunct attachment ambiguity: sentences of the form "NP1 of NP2 C VP", where the relative clause C can attach to either NP2 (structure 1) or NP1 (structure 2). Each sentence is constructed by manipulating lexical content so that only one of the two structures is semantically plausible. We define a target string ("NP2 PP" or "NP2 C") that forms a valid constituent under structure 1 but not under structure 2. The demonstration uses an unambiguous sentence and deletes an NP. If semantic plausibility influences how the model constructs the syntactic structure, the model should delete the target string more frequently under structure 1. We compare the proportion of target string deletions under structure 1 versus under structure 2.

III. EVALUATION WITH GENERAL TESTS

For general tests, we evaluated LLMs from the Qwen family at multiple model sizes: `qwen2.5-flash`, `qwen2.5-plus`, and `qwen2.5-max`. Each test was evaluated over 600 runs per LLM, with each run comprising 24 independent trials. We recorded the model responses and took the difference between the recorded word string and the original test sentence to obtain the deleted word string. We used six prompts (Prompts A–F, see Appendix Table S2; 100 runs per prompt) in total: three adopted from Liu et al. [8] and three newly designed. The remaining experimental settings are provided in Appendix B.

We first analyzed whether LLMs preferred to delete constituents (Test 1). Fig. 1a shows the proportion of each category, averaged across prompts. The constituent rate (i.e., the proportion of single-constituent deletions) was significantly above the chance level of 0.14 for all three LLMs (all $p < .001$,

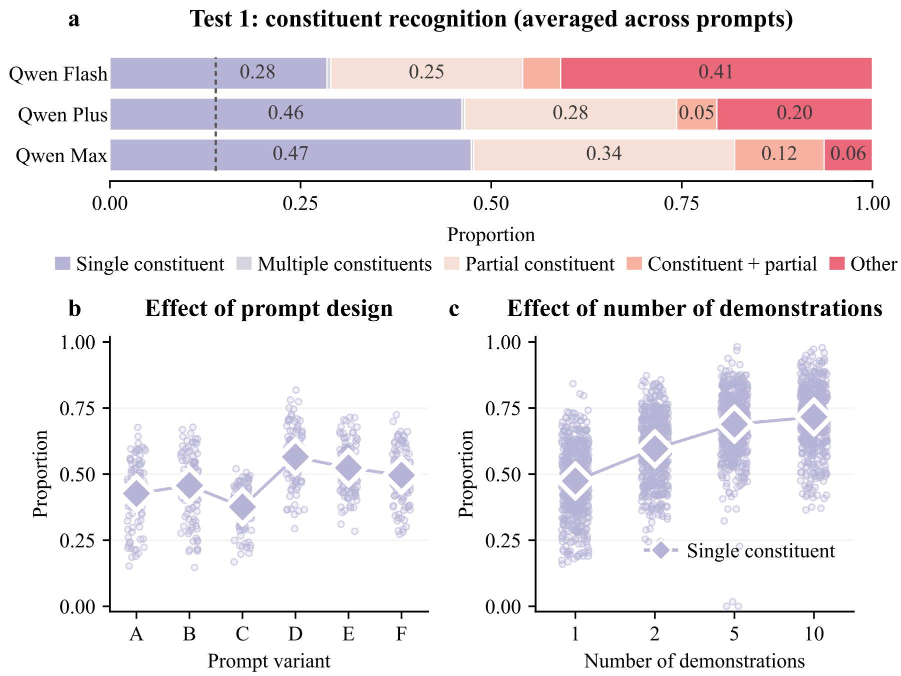

Fig. 1. Results of Test 1. **a.** The proportion of model responses in each category, averaged across prompts. The gray dashed line represents the chance level for single-constituent deletions (see Appendix C for more details). **b.** Constituent rate under different prompts for Qwen Max. **c.** Constituent rate under different numbers of demonstrations for Qwen Max.

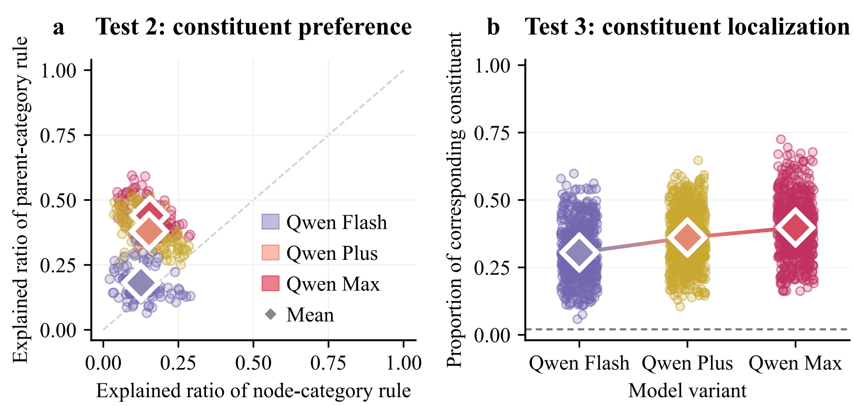

Fig. 2. Results of Tests 2 and 3, averaged across prompts. **a.** Explained ratio of node-/parent-category rule. **b.** The proportion of model responses that deleted the corresponding constituent. The gray dashed line represents the chance level.

one-sided Monte Carlo test; see Appendix C for details of the statistical tests and chance-level calculation), and increased from 0.28 (Qwen Flash) to 0.47 (Qwen Max). Since the constituency trees derived from CoNLL-2000 did not capture all possible constituents, the reported constituent rates might be underestimated. These results suggested that LLMs preferred to delete constituents, and that this preference strengthened as model size increased. The detailed results of Test 1 are shown in Appendix Tables S3 and S4.

This preference varied with prompt design and the number of demonstrations. For Qwen Max, the newly designed prompts (D–F) yielded higher constituent rates than those adopted from Liu et al. [8] (A–C; Fig. 1b), whereas the trend was reversed for Qwen Plus (Appendix Table S3). Constituent rates were therefore sensitive to prompt wording in a model-dependent manner, and WDT-Bench averaged over six prompts rather than reporting single-prompt performance. When multiple demonstrations were provided for a single test sentence, the constituent rate increased steadily, reaching 0.72 with ten demonstrations (Fig. 1c).

We further examined which deletion rules LLMs applied (Test 2). For all LLMs, the explained ratio of the parent-category rule exceeded that of the node-category rule (Fig. 2a), indicating that the LLMs relied on structural position rather than category to delete constituents. This preference for the parent-category rule strengthened as model size increased, becoming most pronounced for Qwen Max. Test 3 investigated whether LLMs could locate and delete the specific constituent that structurally corresponded to the one deleted in the demonstration. Performance rose only modestly with model

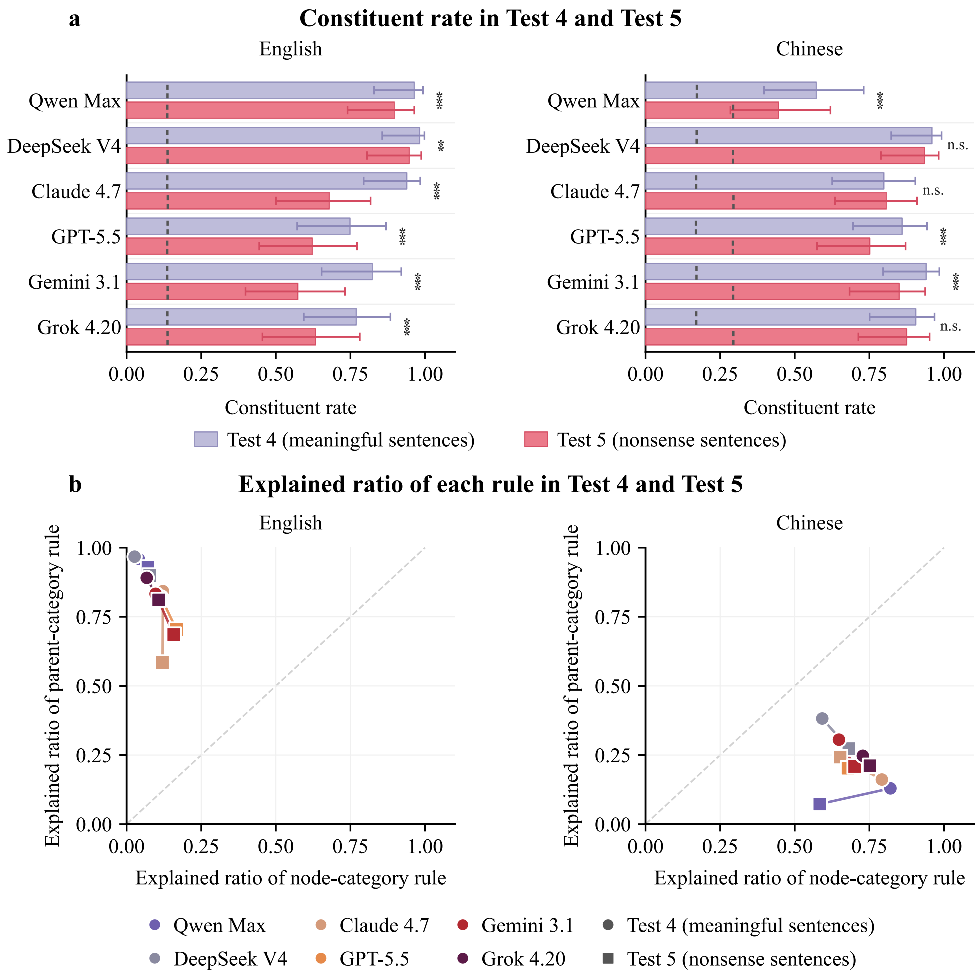

Fig. 3. Results of Tests 4 and 5. **a.** Constituent rate. The gray dashed line represents the chance level. **b.** Explained ratio of node-/parent-category rule. *$p < 0.05$, **$p < 0.01$, ***$p < 0.001$, two-sided permutation test.

scale (Fig. 2b), from 0.31 (Qwen Flash) to 0.40 (Qwen Max), well above the 0.02 chance level of this stricter test. Detailed results of Tests 2 and 3 are shown in Appendix Table S5.

In sum, the general tests demonstrated that WDT-Bench was sensitive to model size and differentiated distinct aspects of general constituent-based processing. All three Qwen models showed evidence of using syntactic structure, yet Qwen Flash could not employ the structure to execute the task reliably.

IV. Evaluation with Diagnostic Tests

While the general tests focused on a single model family (Qwen), the diagnostic tests evaluated a broader range of LLMs to probe specific dimensions of syntactic competence. We evaluated six LLMs: `qwen2.5-max`, `deepseek-v4-pro`, `gpt-5.5`, `claude-opus-4.7`, `gemini-3.1-pro`, and `grok-4.20-0309`. As in the general tests, each run comprised 24 independent trials; each test was evaluated over 30 runs per LLM. We used a single prompt (Prompt A in Appendix Table S2) for diagnostic tests. The remaining experimental settings are provided in Appendix B.

We examined whether the constituent-deletion behavior generalized across languages (Test 4), and how removing semantic content affected the syntactic competence of LLMs (Test 5). Fig. 3a presents the constituent rate of each LLM on meaningful Chinese-English parallel sentences and nonsense sentences. On meaningful sentences, all LLMs achieved relatively high constituent rates in both Chinese and English, indicating that constituent-deletion behavior was not limited to English. However, when content words were replaced with random words, i.e., on nonsense sentences, constituent rates dropped significantly for most LLMs. In English, Claude 4.7 and Gemini 3.1 showed the most pronounced decline; in Chinese, Qwen Max and GPT-5.5 declined most sharply. DeepSeek V4 was a notable exception: it maintained a relatively high constituent rate on nonsense sentences in both languages, suggesting that DeepSeek V4 was less dependent on semantic cues for syntactic processing. These results indicated that for most LLMs, syntactic competence was weakened when semantic content was removed.

In contrast, deletion rule preferences remained stable when content words were randomly replaced. Fig. 3b shows the explained ratio of the node-/parent-category rule in Chinese and English. In English, all LLMs preferred the parent-category rule on both meaningful and nonsense sentences. In Chinese,

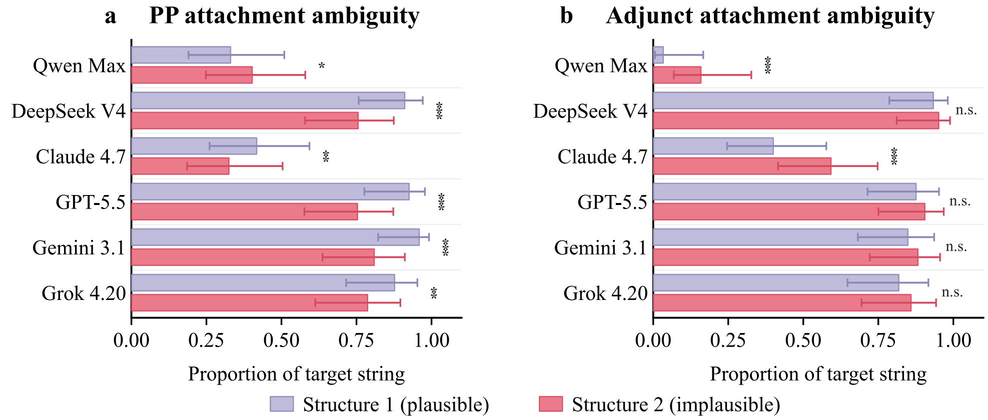

Fig. 4. Results of Test 6. The proportion of model responses that delete the target string for syntactically ambiguous sentences. **a.** PP attachment ambiguity. **b.** Adjunct attachment ambiguity. $^{*}p < 0.05$, $^{**}p < 0.01$, $^{***}p < 0.001$, two-sided permutation test.

however, all LLMs consistently preferred the node-category rule, contrasting with the parent-category rule preference observed in English. This cross-linguistic difference was stable across all six LLMs and both tests, suggesting that LLMs adopted distinct syntactic strategies for Chinese and English.

We further tested whether semantics guided syntactic processing in LLMs when the syntactic structure was ambiguous (Test 6). Fig. 4 presents the proportion of target string deletions under structure 1 versus under structure 2. Since the target string is a semantically plausible constituent only under structure 1, a higher proportion of target string deletions under structure 1 indicates correct integration of semantics into syntactic processing. For PP attachment ambiguity, all LLMs except Qwen Max deleted the target string more frequently under structure 1, indicating that they correctly integrated semantic information. For adjunct attachment ambiguity, the pattern was less consistent across LLMs: Qwen Max and Claude 4.7 deleted the target string more frequently under structure 2, suggesting incorrect syntax-semantics integration, while the remaining LLMs showed no significant difference between structures 1 and 2. Detailed results of Tests 4–6 are shown in Appendix Tables S6 and S7.

In sum, the diagnostic tests demonstrated that WDT-Bench could systematically probe syntactic competence across languages and the interaction with semantics. The cross-linguistic divergence in rule preference (Test 4) was stable across LLMs. For most LLMs, removing semantic content weakened constituent-based processing (Test 5), while semantic plausibility affected constituent processing in PP-attachment ambiguity but showed less consistent effects in adjunct attachment ambiguity (Test 6).

V. Conclusion

In conclusion, we introduced WDT-Bench, an open-source benchmark for implicitly evaluating syntactic competence in LLMs, with a primary focus on constituent-based processing. Using WDT-Bench, we found a clear scaling effect in syntactic processing, a consistent Chinese–English divergence in syntactic strategies, and reduced syntactic processing under semantic disruption for most LLMs, with DeepSeek V4 as a notable exception. These findings highlight the value of implicit evaluation for syntactic competence. WDT-Bench also provides a standardized framework for behavioral comparison across LLMs, facilitating future mechanistic analyses of the identified cross-model differences.

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